Cooperative Game for Carbon Obligation Allocation Among Distribution System Operators to Incentivize the Proliferation of Renewable Energy

Quan Zhou, *Student Member, IEEE*, Mohammad Shahidehpour, *Fellow, IEEE*, Tao Sun, *Student Member, IEEE*, Donghan Feng, *Senior Member, IEEE*, and Mingyu Yan, *Student Member, IEEE*

***Abstract*—The inclusion of electricity consumption in carbon abatement policies can further exploit the carbon reduction potentials in power system operations. In this paper, we propose a strategy for the allocation of carbon obligation as penalties among distribution system operators (DSOs) to incentivize the proliferation of renewable energy. The proposed strategy considers the fairness of the carbon obligation allocation and ensures that DSOs located closer to carbon emitting units would be allocated higher carbon obligations. The interactions among DSOs using the cooperative game theory and the impact of power network topology are comprehensively analyzed in order to properly measure each DSO's contribution to the system carbon obligation. The allocated carbon obligations as cost penalties would incentivize DSOs to accommodate additional renewable generation to reduce the DSO's operation cost. Thus, the proposed allocation strategy provides a technical ground for reducing carbon emissions by dispatching the additional renewable generation and reducing high carbon emission generation. In this paper, Shapley value, Aumann–Shapley rule, and prenucleolus strategies are utilized as three alternatives to allocate carbon obligations among DSOs. Two additional strategies, which are based on existing bus carbon intensity assessments, are also revisited and compared. Relevant allocation problem constraints are presented for evaluating the merits of the proposed strategies. Two case studies are analyzed to highlight the performance of the proposed Shapley value-based strategy in terms of fairness and compatibility for accommodating the additional renewable energy and reducing carbon emissions in power systems.**

***Index Terms*—Carbon obligation allocation, renewable energy, cooperative game theory, distribution system operator.**

Manuscript received October 12, 2018; revised January 24, 2019; accepted March 1, 2019. Date of publication March 7, 2019; date of current version October 30, 2019. Paper no. TSG-01543-2018. *(Corresponding author: Mohammad Shahidehpour.)*

Q. Zhou, M. Shahidehpour, and M. Yan are with the Galvin Center for Electricity Innovation, Illinois Institute of Technology, Chicago, IL 60616 USA (e-mail: qzhou15@hawk.iit.edu; ms@iit.edu; myan9@iit.edu).

T. Sun is with the Department of Civil and Environmental Engineering, Stanford University, Stanford, CA 94305 USA (e-mail: luke18@stanford.edu).

D. Feng is with the Department of Electrical Engineering, Shanghai Jiao Tong University, Shanghai 200240, China (e-mail: donghan.feng.dr@ieee.org).

Digital Object Identifier 10.1109/TSG.2019.2903686

NOMENCLATURE

Index and Sets

i, j	Index of DSOs as market players participating in carbon obligation allocation problem
m, n	Index for buses in power system
k	Index for generators in power system
$\Gamma\_(m)$	Set of buses at branches of bus m.

Parameters

ρ_k	Offer price of generator k
P_{Gk}	Power output of generator k
P_{Di}	Demand of DSO i purchased from the grid
$\mathbf{e}$	Column vector of ones
$\mathbf{SF}$	Shift factor matrix
$\underline{\mathbf{F}}, \bar{\mathbf{F}}$	Vectors of transmission lines' power flow limits
$\underline{\mathbf{P}}_{\mathbf{G}}, \bar{\mathbf{P}}_{\mathbf{G}}$	Vectors of generators' lower and upper power outputs
$\mathbf{u_D}$	Diagonal matrix $diag(u_{D1}, \ldots, u_{Dn_N})$ in which $u_{Di} = 1$denotes that the load in DSO i is supplied; otherwise $u_{Di} = 0$
N	Grand coalition including all the participating DSOs
S, T	Arbitrary non-empty subset of the grand coalition
$c(\cdot)$	Characteristic function of the cooperative game
(N, c)	Formulated cooperative game including the grand coalition N and characteristic function $c(\cdot)$
S^C	Complement set of S in N, i.e., $N \backslash S$
$c'_S(\cdot)$	Characteristic function of the reduced game
$\boldsymbol{\sigma}_{\mathbf{G}}$	Vector of generators' carbon emission rates
$P(\cdot)$	Occurrence probability of a given coalition
n_G	Number of generators in the power system
n_N	Number of DSOs in the grand coalition N
n_S	Number of DSOs in the sub-coalition S
$n_N!$	Number of permutations produced by all the DSOs
$\mathbf{x}, \mathbf{x_1}, \mathbf{x_2}$	Preimputations representing a set of each DSO's allocated carbon obligation $(x_1, x_2, \ldots, x_n)$

$I^0(N, c)$ Preimputation space of the game (N, c) containing all efficient solutions
λ_C Carbon price.

Variables

$\mathbf{P_G}$ Vector of generators' power outputs
$\mathbf{P_D}$ Vector of DSOs' demand
λ Multiplier corresponding to the balance constraint
$\underline{\boldsymbol{\mu}}, \bar{\boldsymbol{\mu}}$ Vector of multipliers corresponding to the power flow limit of transmission lines
$\underline{\boldsymbol{\eta}}, \bar{\boldsymbol{\eta}}$ Vector of multipliers corresponding to the output limits of generators
$\mathbf{P}_{\mathbf{G}}^{(S)}$ Generators dispatch when only DSOs in coalition S are supplied
x_i Allocated carbon obligation of DSO i
$e(S, \mathbf{x})$ Excess of coalition S relative to preimputation $\mathbf{x}$
$x(N)$ Sum of all the DSOs' allocated carbon obligations according to the solution $\mathbf{x}(N, c)$
$x(S)$ Sum of individual DSOs' allocated carbon obligations belonging to coalition S according to the solution $\mathbf{x}(N, c)$
$fci_{f(i)}$ Footprint carbon intensity of the bus DSO i located
$mci_{f(i)}$ Marginal carbon intensity of the bus DSO i located
$\mathbf{CF_n^{gross}}$ Vector of gross bus through-carbon-flows
$\mathbf{CF_G}$ Vector of bus carbon flow injections
$\mathbf{A_u^{gross}}$ Gross upstream distribution matrix
P_{nm} Gross power flow export from bus n at line n-m
P_n Through-power-flow of bus (sum of inflows)

Other notations are defined in the text.

I. Introduction

POWER industry accounts for about 40% of global energy-related CO_2 emissions, making carbon abatement a significant concern. Generally, relevant government policies, incentive options, and mitigation programs focus on carbon emission sources in electricity generation. Among such initiatives, carbon tax and cap-and-trade programs are the two most widely used methods around the world [1]. Carbon tax program has no price volatilities and is stable in revenue. The simplicity of carbon tax makes it easier for implementation. But the political ramifications of introducing additional taxes and the socioeconomic resistance to uncertain global goals for emission reduction are considered as main demerits of carbon tax. Comparatively, the cap-and-trade program promises fix emission reduction levels, which can be easily combined with other policy initiatives. But the cap-and-trade tradeoff is vulnerable to price volatilities of emission allowances and speculations by traders. In these policies, carbon emissions are only viewed as contributing to the environmental cost of electricity generation, which are calculated using fossil fuel consumptions and relevant emission rates.

However, with the development of distributed systems and microgrids, demand-side resources will play essential roles in low-carbon power system operations; so there exist considerable carbon reduction potentials on demand side [1]–[6]. Hence, it is reasonable to assess the corresponding emission reduction potentials and manage the carbon obligation considering electricity consumptions [7]–[13]. In [7], the distinctions between the generation- and the consumption-based accounting principles are demonstrated in countries like Denmark. For industrial and commercial sectors, imposing carbon cost on end-user may perform better than on suppliers in terms of the overall carbon emission reduction [8]. In addition, when the system carbon obligation is allocated to the generation sector, the corresponding carbon cost would be reflected in electricity prices and rolled in electricity purchases. It is viewed that monetizing carbon obligations as penalties in electricity purchases can motivate market participants to take actions for reducing carbon costs in power system operations.

By aggregating distributed resources, flexible demand, and storage, distribution system operators (DSOs) can minimize their operation cost by participating as players in bulk power market operations. With the proliferation of renewable generation and microgrids, DSOs will have an opportunity to supply aggregated loads by electricity purchases from the transmission grid or local renewable generation (e.g., wind and solar photovoltaic) [14], [15]. Since carbon emission rates of renewable generation units are much lower than those of thermal units (e.g., coal and gas-fired units), incentivizing DSOs to accommodate the additional renewable generation as a means of reducing the supply cost can reduce the overall carbon emissions.

In this paper, carbon obligation is monetized and allocated as penalties to DSOs in order to highlight clean renewable generation as a viable alternative for minimizing the total operation cost. In the proposed model, higher carbon obligations are allocated to DSOs which are located closer to thermal generating units with higher emission rates so that any generation dispatch adjustments in such units could have a more profound influence on carbon reduction. In the proposed model, the DSO's participation in electricity markets and higher engagements of renewable energy in electricity supply could have major impacts on carbon abatement efforts and policies.

For allocating carbon obligations among DSOs, two indices have been proposed for measuring the bus carbon intensity, including footprint carbon intensity (FCI) [9]–[11] and marginal carbon intensity (MCI) [12]. The former index estimates the carbon intensity of the DSO's demand using the carbon flow tracing method. The latter index is determined by assessing the impact of locational DSO demand changes on the system carbon obligation through sensitivity analyses. However, the two methods would fail to properly reflect the role of DSO interactions on carbon obligation allocations.

DSOs which participate in power markets will have their demands met simultaneously by electricity generations through the usage of transmission network. So both generators and transmission networks are treated as common resources, which are shared by all the participating DSOs and are utilized for satisfying DSOs' demands. Thus, all the participating DSOs form a cooperative relationship in terms of system carbon obligation. Therefore, we apply a cooperative game method to allocate carbon obligations among DSOs. Here, the carbon obligation allocation is viewed as a cost allocation problem in power systems. The cooperative game theory solutions offer a good performance in power systems in terms of fairness, efficiency and stability for addressing cost allocations issues including those of transmission service costs [16]–[18], planning expansion [19], [20], network losses [21], [22], energy storage resources [23], [24], demand-side resources [25], [26], start-up costs [27], firm-energy rights [28], renewable energy portfolio [29], etc.

The main contributions of this paper are listed as follows:

a) Electricity consumption should be included in the carbon abatement policies. The carbon obligation imposed on demand side encourages the accommodation of distributed renewable energy and the participation of demand-side resources in system carbon abatement. Unlike the direct relationship between electricity generation and carbon emission, the relationship between electricity consumption and carbon emission is indirect which should be evaluated carefully for fully exploiting the carbon abatement potentials in demand side;
b) Both generators and transmission networks are treated as common resources for satisfying DSO demands, which form a cooperative relationship in terms of system carbon obligation. Thus, the carbon obligation allocation among DSOs is formulated as a cooperative game with a transferable utility. The set of players corresponds to the DSOs participating in the power market. The characteristic function corresponds to the system carbon obligation. The allocation problem is to allocate the system carbon obligation among players in the grand coalition;
c) The prenucleolus, Shapley value, and Aumann-Shapley rule are utilized as three viable solutions for allocating carbon obligations among DSOs. The corresponding allocation solutions are also compared with those of the two existing carbon intensity assessment methods (i.e., FCI and MCI). Furthermore, additional cooperative game constraints are considered to assess these feasible solutions. It is demonstrated that the Shapley value-based solution meets the prevailing constraints while other solutions violate certain constraints. In addition, the Aumann-Shapley rule provides an alternative solution considering a large number of market players.

The remainder of this paper is organized as follows. In Section II, the carbon obligation allocation among DSOs is discussed and formulated as a cooperative game method. In Section III, relevant constraints of the allocation problem are formulated in terms of cooperative game applications. In Section IV, allocation solutions using different methods are presented and the results are discussed and compared according to the formulated constraints. Case studies are presented in Section V to validate the effectiveness of the proposed carbon obligation allocation strategy. Section VI concludes the paper.

II. Carbon Obligation Allocation Among DSOs in Cooperative Game

Here, participating DSOs in a bulk electricity market form a cooperative relationship in terms of their contributions to system carbon obligation. In a cooperative game, participating DSOs as players form the grand coalition N in the ISO's market. The ISO's security-constrained economic dispatch (SCED) is expressed as:

$$\begin{aligned} &Min \ \sum_{k=1}^{n_G} \rho_k P_{Gk} \\ &s.t. \ \mathbf{e}^T\mathbf{P_G} - \mathbf{e}^T\mathbf{u_D}\mathbf{P_D} = 0 && \to \lambda \\ &\quad \underline{\mathbf{F}} \leq \mathbf{SF}(\mathbf{P_G} - \mathbf{u_D}\mathbf{P_D}) \leq \bar{\mathbf{F}} && \to \underline{\boldsymbol{\mu}}, \bar{\boldsymbol{\mu}} \\ &\quad \underline{\mathbf{P}}_\mathbf{G} \leq \mathbf{P_G} \leq \bar{\mathbf{P}}_\mathbf{G} && \to \underline{\boldsymbol{\eta}}, \bar{\boldsymbol{\eta}} \end{aligned} \tag{1}$$

Assume S is an arbitrary subset of N as a possible coalition. The system carbon obligation of coalition S is calculated using the characteristic function as $c(S) = \boldsymbol{\sigma}_\mathbf{G}\mathbf{P}_\mathbf{G}^{(S)}$, where the generation dispatch $\mathbf{P}_\mathbf{G}^{(S)}$ is obtained by solving (1) when DSOs belonging to S are only supplied (i.e., $u_{Di} = 1, \forall i \in S$). Accordingly, the carbon obligation allocation problem turns into determining an appropriate solution in the cooperative game (N, c) with N players and a characteristic function $c(\cdot)$. Noting that the SCED in (1) presented as a linear programming problem might have multiple optimal solutions. Although these dispatch solutions have the same optimal purchasing cost, they might result in different system carbon obligations according to the characteristic function. So, we select the optimal dispatch solution among the candidates by minimizing the system carbon obligation.

The system carbon obligation for supplying all DSOs simultaneously may not be the same as the sum of system carbon obligations for supplying each DSO individually. Assume there are n_N DSOs in the bulk power market and DSO i is the only one supplied (i.e., $u_{Di} = 1$ *and* $u_{Dj} = 0$ for $\forall j \neq i$); then the corresponding system carbon obligation is $c(i) = \boldsymbol{\sigma}_\mathbf{G}\mathbf{P}_\mathbf{G}^{(i)}$. When all DSOs are supplied simultaneously (i.e., $u_{Di} = 1$ for $\forall i \in N$), the system carbon obligation which is denoted by $c(N) = \boldsymbol{\sigma}_\mathbf{G}\mathbf{P}_\mathbf{G}^{(N)}$ is allocated among participating DSOs. Then, $c(N) \neq \sum_{i \in N} c(i)$, noting that the characteristic function $c(\cdot)$ is not necessarily concave. Thus, the subadditivity cannot always hold in this cooperative game. Similarly, the subadditivity of the cooperative game in transmission loss allocation is not satisfied due to the multiplicity of loads [22].

III. Carbon Obligation Allocation Constraints

In this section, cooperative game constraints are illustrated considering the characteristics of the carbon obligation allocation problem. When certain constraints are not satisfied, the carbon obligation allocation performance might be deemed as unfair in our proposed solution.

A. Constraints

1) Efficiency (EFF): The sum of DSOs' allocated carbon obligations is equal to the power system's carbon emission,

$$x(N) = c(N) \tag{2}$$

This constraint ensures that the system carbon obligation is fully covered by the allocation solution.

2) Reasonability (REA): Each DSO's carbon obligation allocation will not exceed its maximum marginal contribution to the system carbon obligation in any coalition and will not be less than its minimum marginal contribution,

$$\min(c(S\cup i) - c(S)) \leq x_i \leq \max(c(S\cup i) - c(S)) \tag{3}$$

This constraint ensures that the allocation solution is within an appropriate range and be reasonable from both sides.

3) Symmetry (SYM): DSOs i and j are defined as symmetrical players if they have the same marginal contributions to every coalition that does not include the players i and j,

$$c(S\cup i) = c(S\cup j), \forall S \subseteq N\backslash\{i, j\} \tag{4}$$

Accordingly, carbon obligation allocations would be identical,

$$x_i(N, c) = x_j(N, c) \tag{5}$$

This constraint ensures that the symmetrical players are treated equally in the carbon allocation solution.

4) Balanced Contributions (BC): The changes in carbon obligation allocation caused by a player's withdrawal from the coalition would be the same for any two DSOs i and j in the grand coalition,

$$x_i\left(N\backslash j, c^{N\backslash j}\right) - x_i(N, c) = x_j\left(N\backslash i, c^{N\backslash i}\right) - x_j(N, c) \tag{6}$$

Otherwise, there will be multiple options for carbon obligation allocation between any two DSOs, which might lead to contradictory allocation solutions.

5) Davis-Maschler Reduced Game Consistency (DMRGC): The DM reduced game with respect to S and $\mathbf{x}$ is denoted by $(S, c_{S,x})$ [30] with a characteristic function expressed as

$$c_{S,x}(T) = \begin{cases} 0, & T = \phi \\ x(S), & T = S \\ \min\limits_{Q\subseteq S^C}[c(T\cup Q) - x(Q)], & else \end{cases} \tag{7}$$

where T represents an arbitrary subset of S, Q represents an arbitrary subset of $S^C = N\backslash S$, and $x(Q) = \sum_{i\in Q} x_i$ represents the DSO's sum of allocated carbon obligations pertaining to Q in the game (N, c).

In the DM reduced game $(S, c_{S,x})$, the coalition T members continue to cooperate with selected members of S^C, and $c_{S,x}(T)$ is the total carbon obligation in the coalition T. The DMRGC requires that the same carbon obligations as in the original game to be allocated to DSOs when a DM reduced game is considered,

$$x_i\big(S, c_{S,x}\big) = x_i(N, c) \quad \forall i \in S. \tag{8}$$

6) Hart-Mas-Colell Reduced Game Consistency (HMRGC): The characteristic function of the HM reduced game $\left(S, c'_{S,x}\right)$ is defined as [31],

$$c'_{S,x}(T) = c\left(T\cup S^C\right) - \sum_{i\in S^C} x_i\left(T\cup S^C, c\right) \tag{9}$$

where $\sum_{i\in S^C} x_i(T\cup S^C, c)$ represents the sum of all DSOs' carbon obligations in S^C considering the reduced game $(T\cup S^C, c)$.

In the HM reduced game $(S, c'_{S,x})$, all members of the T coalition cooperate with the coalition S^C when $S\backslash T$ is not present. For $T \in \{\phi, S\}$, $c'_{S,x}(T)$ coincides with $c_{S,x}(T)$ [32]. The HMRGC ensures that the solution for any HM reduced game is consistent with the allocation solution in the original game,

$$x_i\big(S, c'_{S,x}\big) = x_i(N, c) \quad \forall i \in S \tag{10}$$

The HM reduced game differs from the Davis-Maschler (DM) reduced game in the following two respects. First, DSOs in T as the subset of S cannot freely choose partners from S^C; i.e., their cooperation is stuck with S^C. Second, the DSOs' carbon obligations in S^C are assigned in accordance with the allocation of the $T\cup S^C$ coalition, rather than that of the grand coalition N. Accordingly, the HM reduced game fits the situation involving a neutral third party evaluator, while the DM form is more suitable for reallocations by participants [32].

Considering a multi-area power system and a set S of DSOs in one area, the remaining set of DSOs is denoted as S^C. For any subset $T \subset S$, the DSOs in T calculate their assigned carbon obligation locally assuming that they are the only participants in S. Thus, the carbon obligation assigned to the coalition T is considered in a reduced game $(T\cup S^C, c)$ given in (9). Accordingly, the HMRGC constraint assures that the local solution calculated by any coalition S is no different than the solution calculated by N centrally.

B. Discussions on Constraints in Carbon Allocation

The efficiency constraint ensures that the allocation solution covers the system carbon obligation, determining whether the allocation solution can be directly applied. The constraints including reasonability, symmetry, and BC focus on the fairness of the allocation solution, ensuring that each DSO's contribution to the system carbon obligation is measured properly. In power system market operations, DSOs are required to supply individual loads and cannot be withdrawn easily from the market. Thus, the HM reduced game models this requirement more precisely than the DM reduced game. In our allocation problem, we apply HMRGC rather than DMRGC as a constraint to ensure that the allocation solution can remain the same no matter whether the solution is calculated locally or centrally.

IV. Carbon Obligation Allocation Methods

There are five allocation methods discussed and compared in this section. Three cooperative game theory-based methods, including prenucleolus, Shapley value, and Aumann-Shapley rules are utilized to allocate the system carbon obligation among DSOs. For comparison, the carbon flow tracing method [9]–[11] and the marginal method [12], [13] are also considered.

A. Prenucleolus

The nucleolus with respect to the set of preimputations is called prenucleolus [33]. That is, nucleolus is constrained by the nonempty core while prenucleolus is not. Since the subadditivity of the formulated cooperative game cannot always hold, we use the prenucleolus rather than the nucleolus to allocate the carbon obligation. The key notion which underlies nucleolus and prenucleolus is excess, which is defined as

$$e(S, \mathbf{x}) = x(S) - c(S), \mathbf{x} \in I^0(N, c) \tag{11}$$

Excess is interpreted as the dissatisfaction of a particular coalition with the preimputation $\mathbf{x}$. Prenucleolus is to minimize all dissatisfactions in the lexicographic order for obtaining a unique preimputation. The prenucleolus-based allocation will minimize the largest dissatisfaction consistently. Thus, prenucleolus is solved using the following procedure.

1) Minimize the Maximal Excess: The minimization of the maximal excess is formulated as

$$\begin{aligned} \min \quad & \varepsilon \\ s.t. \quad & e(S, \mathbf{x}) \le \varepsilon \quad \forall S \subseteq N \\ & x(N) = c(N) \end{aligned} \tag{12}$$

The maximal excess is denoted by θ_1. The solutions which minimize the maximal excess are expressed as

$$\mathbf{x_1} = \left\{ \mathbf{x} \in I^0(N, c) | e(S, \mathbf{x}) \le \theta_1, \forall S \subseteq N \right\} \tag{13}$$

The corresponding coalition with the maximal excess θ_1 is denoted by

$$\Sigma_1 = \{S \subseteq N | e(S, \mathbf{x}) = \theta_1, \quad \forall \mathbf{x} \in \mathbf{x}_1\} \tag{14}$$

If $\mathbf{x_1}$ is unique, it represents the prenucleolus of the game. Otherwise, consider the second maximal excess.

2) Minimize the Second Maximal Excess: The linear minimization of the second maximal excess is formulated as

$$\begin{aligned} \min \quad & \varepsilon \\ s.t. \quad & e(S, \mathbf{x}) = \theta_1, \quad \forall S \subseteq \Sigma_1 \\ & e(S, \mathbf{x}) \le \varepsilon, \quad \forall S \not\subset \Sigma_1 \\ & x(N) = c(N) \end{aligned} \tag{15}$$

The solution is denoted by θ_2 and solutions which minimize the second maximal excess are expressed as $\mathbf{x_2}$. If $\mathbf{x_2}$ is unique, it is the prenucleolus of the game. Otherwise, we continue the procedure with the third maximal excess, and repeat the procedure until the unique preimputation is obtained. The prenucleolus is viewed as the lexicographic least-core in the core when the core is nonempty; otherwise, it exists even if the core is empty [33]. The unique prenucleolus is the corresponding solution for carbon obligation allocation among DSOs.

B. Shapley Value

Shapley value would not emphasize strategic considerations and focus on analyzing the marginal contributions of each player to possible coalitions [34]. Unlike the partial assessment of cost causation in marginal cost analyses, the Shapley value considers all players' costs in a holistic perspective. If there are n_N DSOs participating in a power market, the number of all possible permutations is $n_N!$ and the number of non-empty coalition would be $2^{n_N} - 1$. In Shapley value, each market player has the same opportunity to be in any permutation position where players are treated equally in a fair and unbiased process as the entry order is eliminated [21].

Shapley value allocates to each DSO the average of its marginal contributions over all permutations of N, in which case the permutations have the same probability. Accordingly, the corresponding carbon obligation of each DSO is defined as

$$x_i = \sum_{S \subseteq N \setminus i} P(S)(c(S \cup i) - c(S)) \tag{16}$$

where $c(S \cup i) - c(S)$ represents the incremental carbon obligation for including DSO i in the coalition S.

The probability of DSOs in coalition S preceding DSO i in a random permutation of N is $n_S!(n_N - n_S - 1)!/n_N!$. Accordingly, the allocated carbon obligation of DSO i using the Shapley value is calculated as

$$x_i = \sum_{S \subseteq N \setminus i} \frac{n_S!(n_N - n_S - 1)!}{n_N!}[c(S \cup i) - c(S)] \tag{17}$$

According to (17), the calculation of the Shapley value-based solution requires $2^{n_N} - 1$ SCED solutions in (1). However, the computational complexity of Shapley value becomes exceptionally high with the increasing number of market players as the number of nonempty coalitions grows exponentially.

C. Aumann-Shapley Rule

Considering the limitation of the Shapley value in computational feasibility involved with a larger number of market players, the Aumann-Shapley rule has long been viewed as the continuous generalization of the Shapley value to deal with cost allocations. The Aumann-Shapley value is a natural combination of average and marginal cost sharing rules. Each player's value is its marginal contributions averaged over all vectors $t\mathbf{P_D}$, $0 \le t \le 1$, which are defined as the rays from $\mathbf{0}$ to $\mathbf{P_D}$ [32]. Thus, the corresponding carbon obligation allocation is calculated as:

$$x_i = P_{Di} \int_0^1 \frac{\partial c(t\mathbf{P_D})}{\partial P_{Di}} dt \tag{18}$$

Accordingly, the Aumann-Shapley rule-based solution is the average of DSOs' marginal costs (i.e., marginal contributions in the system carbon obligation) when their demands increase uniformly from zero to their current values [16], [28]. Thus, small changes in DSOs' demand will not lead to large changes in the Aumann-Shapley rule-based solution. However, the numerical calculation of (18) is achieved by discretizing t into n segments to represent the characteristic function $c(\cdot)$ which is non-differentiable with respect to P_{Di}. Thus, the Aumann-Shapley rule-based solution is calculated using the following discrete form as:

$$x_i = P_{Di} \cdot \sum_{l=1}^{n} \frac{1}{n} \cdot \frac{c(l\boldsymbol{P}/n + \Delta P_{Di}) - c(l\boldsymbol{P}/n)}{\Delta P_{Di}} \tag{19}$$

where ΔP_{Di} represents a small increment in the demand of DSO i, and $[c(l\boldsymbol{P}/n + \Delta P_{Di}) - c(l\boldsymbol{P}/n)]/\Delta P_{Di}$ represents the change in the system carbon obligation with respect to per unit increment in the demand of DSO i in the lth segment.

D. Footprint Carbon Intensity

The carbon flow in power networks and the corresponding calculating method were introduced in [9]. By adding carbon label to power flows, carbon flow is defined as a virtual network flow from generation to consumption. The basic assumptions applied in the carbon flow tracing method are similar to those of the power flow tracing, i.e., the proportional-sharing rule and the priority of generators to supply the demands at the same bus [9], [10]. Wang *et al.* [15] proposes a novel standardized matrix modeling method for multi-energy systems. The method is able to model the energy flow in a linear manner so that the analysis of multi-energy systems can be easily automated. A carbon flow tracing method across different energy systems is further proposed based this matrix modeling method [11].

Using the carbon flow tracing method and the power flow tracing method, the lossy carbon flow will be traced in a power network by $\mathbf{CF_n^{gross}} = (\mathbf{A_u^{gross}})^{-1}\mathbf{CF_G}$ in which any element of $\mathbf{A_u^{gross}}$ is defined as

$$\left[\mathbf{A_u^{gross}}\right]_{mn} = \begin{cases} 1 & m = n \\ -|P_{nm}|/P_n & n \in \Gamma_-(m) \\ 0 & else \end{cases} \tag{20}$$

The footprint carbon intensities (FCIs) vector for each DSO is denoted as $\mathbf{FCI} = \mathbf{CF_n^{gross}}/\mathbf{P_n}$. FCI, which is measured in tons of CO_2/MWh in a particular bus, represents the carbon footprint per unit of the bus power. Accordingly, the carbon obligation allocated to a DSO is expressed as $x_i = P_{Di} \cdot fci_{f(i)}$.

E. Marginal Carbon Intensity

Marginal carbon intensity (MCI) represents the impact of locational carbon footprint in constrained transmission power networks. MCI at DSO i ($MCI_i = \partial C/\partial P_{Di}$), measured in tons of CO_2/MWh, is defined as the incremental change in the system carbon obligation with respect to a small change in the demand of DSO i. Combining with locational marginal prices (LMPs), the MCI vector in a lossless power network is expressed by [12]

$$\mathbf{MCI} = \frac{\partial \mathbf{LMP}}{\partial \lambda_C} = \frac{\partial \lambda}{\partial \lambda_C} - \frac{\partial \mathbf{SF}^T\left(\bar{\boldsymbol{\mu}} - \underline{\boldsymbol{\mu}}\right)}{\partial \lambda_C} \tag{21}$$

According to the MCI definition, the MCI-based solution is obtained as:

$$x_i = P_{Di} \cdot \frac{c(\boldsymbol{P} + \Delta P_{Di}) - c(\boldsymbol{P})}{\Delta P_{Di}} \tag{22}$$

Compared to (20), the MCI-based solution only considers each DSO's marginal contribution in terms of the system carbon obligation when the DSOs' demands are given as the current values. Any small increment of DSO's demand will be supplied by marginal generators. Considering a transmission-constrained power system, there might be multiple marginal generators with different CO_2 rates when the transmission system is congested [35]–[37]. The MCI of each DSO is a linear combination of CO_2 rates of marginal generators, which might be higher than the highest CO_2 rate of participating generators, or can even be negative. Thus, a small modification in certain DSOs' demands might influence the MCI-based solution drastically, implying that the MCI-based allocation method is vulnerable to the misbehaviors of participants.

TABLE I
COMPARISON OF DIFFERENT ALLOCATION SOLUTIONS

Constraints	Prenucleolus	SV	AS	FCI	MCI	Proportional
EFF	+	+	+	+	-	+
REA	+	+	+	+	-	-
SYM	+	+	+	-	-	-
BC	-	+	-	-	-	-
HMRGC	-	+	-	-	-	-

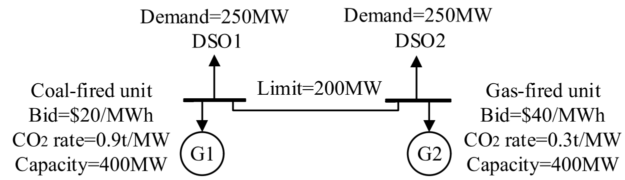

Fig. 1. Two-bus power system.

F. Comparison of Different Allocation Solutions

The comparison of different allocation solutions is presented in Table I in which "+" means that the constraint is satisfied in the allocation solution and "–" indicates otherwise. The proportional principle-based solution, which assigns the carbon obligation according to a DSO's purchased electricity, is also listed for comparison. The Shapley value-based allocation solution, which considers DSO interactions and relevant factors such as the network topology, meets all the constraints. Comparatively, the other methods fail to properly measure each DSO's contribution to the system carbon obligation and thereby violate some of the constraints.

However, to a large-scale system, the computational complexity of the Shapley value or prenucleolus becomes extremely high because the number of nonempty coalitions grows exponentially according to the number of participants. Thus, the Aumann-Shapley rule offers an alternative to the Shapley value-based solution when many players participate in a market, though it does not satisfy the BC and the HMRGC constraints.

V. CASE STUDIES

In this section, the five allocation methods (i.e., Shapley value, Aumann-Shapley rule, prenucleolus, FCI, and MCI) are tested using three test systems: two-bus power system, PJM-five-bus power system and IEEE-30-bus power system. Without the loss of generality, the three cases are considered under SCED in (1).

A. Case Study 1: Two-Bus Power System

The two-bus power system is shown in Fig. 1, which has two generators, two DSOs and one transmission line.

TABLE II
SYSTEM CARBON OBLIGATIONS AND GENERATION DISPATCH

Coalitions	{DSO1}	{DSO2}	{DSO1,DSO2}
Carbon obligation(t/h)	225	225	390
G1 (MW)	250	250	400
G2 (MW)	0	0	100

TABLE III
CARBON OBLIGATION ALLOCATIONS (T CO_2)

Allocation solutions	x_1	x_2	Total
Shapley value	195	195	390
Aumann-Shapley	195	195	390
Prenucleolus	195	195	390
FCI	225	165	390
MCI	75	75	150
Proportional	195	195	390

TABLE IV
SYSTEM CARBON OBLIGATIONS AND GENERATION DISPATCH

Coalitions	{DSO1}	{DSO2}	{DSO1,DSO2}
Carbon obligation(t/h)	225	195	390
G1 (MW)	250	200	400
G2 (MW)	0	50	100

TABLE V
CARBON OBLIGATION ALLOCATIONS WITH TRANSMISSION LIMIT (T CO_2)

Allocation solutions	x_1	x_2	Total
Shapley value	210	180	390
Aumann-Shapley	195	195	390
Prenucleolus	210	180	390
FCI	225	165	390
MCI	75	75	150
Proportional	195	195	390

1) Relaxing Transmission Line 1-2 Constraint: The two independent DSOs are considered as two players, in which there are three nonempty coalitions: {DSO1}, {DSO2} and {DSO1, DSO2}. The coalitions' carbon obligations and their corresponding generator dispatch are shown in Table II. Accordingly, the carbon reduction is $225 + 225 - 390 = 60$ tons, which includes the dispatch of generator G2 with a lower CO_2 rate but a higher electricity price. The generator G2 dispatch is positive because the total demand is larger than the generator G1 capacity when line 1-2 is not constrained. According to (4) and (5), DSO1 and DSO2 should be allocated equal carbon obligations because they are symmetrical players with the same marginal contributions to coalitions which do not include them, i.e.,

$$\begin{cases} c(\{\text{DSO1}\}) - c(\phi) = c(\{\text{DSO2}\}) - c(\phi) = 225t \\ c(N) - c(\{\text{DSO2}\}) = c(N) - c(\{\text{DSO1}\}) = 165t \end{cases} \tag{23}$$

In Table III, different methods are applied to allocate the system carbon obligation among the two DSOs. The proportional principle-based solution is given for comparison. By applying the Shapley value, Aumann-Shapley rule, or prenucleolus, the carbon reduction is equally divided between DSO1 and DSO2, which satisfies the symmetry constraint. According to FCI, the dispatch of generator G2 is only consumed by DSO2 and hence the carbon reduction is all allocated to DSO2, which violates the symmetry constraint. For the MCI-based solution, the system carbon obligation is not fully covered by the allocated carbon obligations among two the DSOs, and then the efficiency constraint is violated. By the proportional principle, the carbon obligation is assigned according to each DSO's demand.

2) Considering the Constrained Transmission Line 1-2: The carbon obligation of coalition {DSO2} and the corresponding dispatch of generators are changed when the transmission line 1-2 is constrained. DSO1 and DSO2 are no longer symmetrical players in Table IV and different allocation solutions are shown in Table V. The results remain the same with respect to the Aumann-Shapley rule, FCI, MCI, and proportional principle. However, the carbon reduction has changed to $225 + 195 - 390 = 30$ tons because 50MW of the generator G2 dispatch is attributed to DSO2 when the transmission line is constrained.

Although the Aumann-Shapley rule could satisfy the efficiency and reasonability constraints, it violates the BC constraint and thereby fails to recognize that the carbon reduction is changed. The change in the carbon reduction is recognized by the Shapley value or prenucleolus, which is equally divided between DSO1 and DSO2. The Shapley value coincides with the prenucleolus on the class of two-player allocation problems as shown in Tables III and V. However, this conclusion may not be generalized to n-player (n>2) allocation problems.

The allocated carbon obligation of DSO1 is larger than that of DSO2. Then the DSO1 would be more motivated to reduce its thermal demand by accommodating renewable generation. Furthermore, assume a 150MW of local renewable generation in DSO1, which reduces the dispatch of coal-fired unit G1 and the gas-fired unit G2 by 100 and 50MW, respectively. Although a lower DSO thermal demand (motivated by the Shapley-value allocation strategy) reduces the dispatch of the two thermal units, the unit G1 dispatch with a higher CO_2 rate is decreased more significantly.

B. Case Study 2: PJM-Five-Bus Power System

The PJM-five-bus power system is shown in Fig. 2 with the generator data given in Table VI. The data for transmission lines and DSO demands are given in [38]. The grand coalition including the three DSOs in the game is represented by N = {B,C,D}, which has six nonempty subsets: {B}, {C}, {D}, {B,C}, {B,D}, {C,D}. The system carbon obligations for different coalitions and the corresponding dispatches of generators are shown in Table VII.

In Table VIII, the carbon obligation allocations are calculated using different allocation methods discussed in Section IV. The proportional principle-based allocation solution is given for comparison. The demands of DSOs B and D are the same and carbon obligations of coalitions {B} and {C} are almost the same. When including DSO D into the

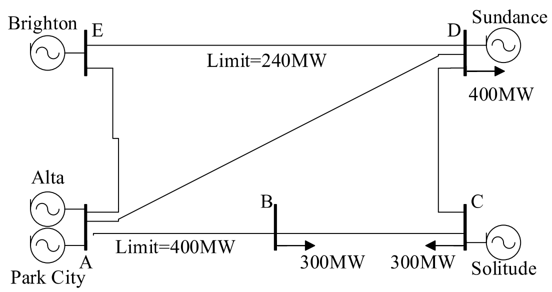

Fig. 2. PJM-five-bus power system.

TABLE VI
GENERATION DATA FOR PJM-FIVE-BUS POWER SYSTEM

	Alta	Park City	Solitude	Sundance	Brighton
Bus	A	A	C	D	E
Energy type	Hydro	Hydro	Gas	Gas	Coal
Offer Price ($/MW)	14.00	15.00	30.00	40.00	20.00
CO_2 rate (t/MW)	0.00	0.00	0.30	0.30	0.90
Capacity (MW)	40.00	130.00	520.00	200.00	600.00

TABLE VII
SYSTEM CARBON OBLIGATIONS (T CO_2) AND GENERATION DISPATCH (MW)

Coaltions	{B}	{C}	{D}	{B,C}	{B,D}	{C,D}	{B,C,D}
Obligation	82.34	82.41	172.48	356.15	429.97	399.63	519.97
Alta	40.00	40.00	40.00	40.00	40.00	40.00	40.00
Park City	170.00	170.00	170.00	170.00	170.00	170.00	170.00
Solitude	0	0	0	0	25.34	76.10	325.34
Sundance	0	0	0	0	0	0	0
Brighton	91.48	91.57	191.65	395.72	469.29	418.66	469.29

TABLE VIII
CARBON OBLIGATION ALLOCATIONS (T CO_2)

Allocation Methods	x_B	x_C	x_D	Sum
Shapley Value	156.10	140.96	222.91	519.97
Aumann-Shapley	182.17	167.89	169.91	519.97
Prenucleolus	143.25	143.32	233.39	519.97
FCI	134.76	99.08	286.13	519.97
MCI	121.69	90.00	-0.02	211.67
Proportional	155.99	155.99	207.99	519.97

two coalitions, the carbon obligations of the two new coalitions {B,D} and {C,D} are different for the same total demand due to DSOs' locational factors. Applying power transfer distribution factor (PTDF), we find that $\mathrm{PTDF}_{\text{E-D,E-B}} < \mathrm{PTDF}_{\text{E-D,E-C}}$, where $\mathrm{PTDF}_{\text{E-D,E-B}}$ is the flow change in line E-D when one unit of power is transferred from bus E to bus B. Thus, the Brighton dispatch is more limited for transfer to bus C than bus B when the transmission line E-D is congested. For both coalitions {C,D} and {B,D}, the transmission line E-D is congested. Accordingly, Solitude, which has a lower CO_2 rate than that of Brighton, is dispatched more in coalition {C,D} as compared with that in coalition {B,D}. The carbon obligation of coalition {B,D} is larger than that of coalition{C,D} though they have the same total demand. Therefore, the assigned carbon obligation of DSO B should be larger than that of DSO C though they have the same demand.

Applying the efficiency constraint, the sum of allocated carbon obligations of the three DSOs should cover the total

TABLE IX
DSOS' INTERACTIONS MEASURED BY ALLOCATION METHODS (T CO_2)

Interaction between DSOs B and C						
Allocation Methods	x'_B	x_B	$x'_B - x_B$	x'_C	x_C	$x'_C - x_C$
SV	169.91	156.10	13.81	154.78	140.96	13.81
AS	184.27	182.17	2.10	171.27	167.89	3.38
Prenucleolus	169.91	143.25	26.66	154.78	143.32	11.45
FCI	143.84	134.76	9.08	126.17	99.08	27.09
MCI	121.69	121.69	0.00	90.00	90.00	0.00
Proportional	184.27	155.99	28.28	171.27	155.99	15.28
Interaction between DSOs B and D						
Allocation Methods	x''_B	x_B	$x''_B - x_B$	x''_D	x_D	$x''_D - x_D$
SV	178.04	156.10	21.94	244.85	222.91	21.94
AS	178.08	182.17	-4.10	228.36	169.91	58.45
Prenucleolus	178.04	143.25	34.79	244.85	233.39	11.45
FCI	145.83	134.76	11.07	273.12	286.13	-13.01
MCI	275.27	121.69	153.58	0.14	-0.02	0.16
Proportional	178.07	155.99	22.08	228.36	207.99	20.37
Interactions between DSOs C and D						
Allocation Methods	x'''_C	x_C	$x'''_C - x_C$	x'''_D	x_D	$x'''_D - x_D$
SV	178.11	140.96	37.15	260.06	222.91	37.15
AS	178.08	167.89	10.19	245.70	169.91	75.79
Prenucleolus	178.11	143.32	34.79	260.06	233.39	26.66
FCI	210.31	99.08	111.24	286.13	286.13	0.00
MCI	275.51	90.00	185.51	-0.02	-0.02	0.00
Proportional	178.07	155.99	22.08	245.69	207.99	37.71

system carbon obligation, 519.97 tons. In Table VIII, only the MCI-based solution fails to satisfy the efficiency constraint.

Applying the reasonability constraint, the solution should be within 82.34 tons $\leq x_B \leq$ 273.74 tons, 82.41 tons $\leq x_C \leq$ 273.81 tons, and 163.82 tons $\leq x_D \leq$ 347.63 tons. The allocations satisfy the constraint except the MCI-based solution which results in a negative allocated carbon obligation to DSO D.

Applying the BC constraint, interactions between every two DSOs measured by these allocation methods are shown in Table IX. Taking the interaction between DSOs B and C for example, x'_B represents the assigned carbon obligation of DSO B after the withdrawal of DSO C, and x'_C represents the assigned carbon obligation of DSO C after the withdrawal of DSO B. According to (6), the impact of the withdrawal of DSO C on the assigned carbon obligation of DSO B ($x'_B - x_B$) should be the same as that of the withdrawal of DSO B on the assigned carbon obligation of DSO C ($x'_C - x_C$). In Table IX, only the Shapley value-based allocation solution satisfies the BC constraint.

Applying the HMRGC, assume DSOs B and C calculate their allocated carbon obligation locally. According to (9), the system carbon obligation of each coalition in the HM reduced game is calculated as

$$\begin{cases} c'_S(B) = c(B \cup D) - x_D(B \cup D, c) = 169.92t \\ c'_S(C) = c(C \cup D) - x_D(C \cup D, c) = 154.78t \\ c'_S(B, C) = c(B, C \cup D) - x_D(B, C \cup D, c) = 297.06t \end{cases} \tag{24}$$

Thus, the locally calculated allocation solution for DSOs B and C are denoted by

$$\begin{cases} x'_{S,B} = \left[c'_S(B, C) + c'_S(B) - c'_S(C)\right]/2 = 156.10t \\ x'_{S,C} = \left[c'_S(B, C) + c'_S(C) - c'_S(B)\right]/2 = 140.96t \end{cases} \tag{25}$$

In Table VIII, only the Shapley value-based solution matches the locally calculated allocations of B and C in (24), which satisfies the HMRGC constraint. Comparatively, for other allocation solutions, the locally calculated allocations are different from the centrally calculated allocations, which violate the HMRGC constraint.

Using the Aumann-Shapley rule, B is additionally supplied by Brighton with a higher CO_2 rate, as we increase the three DSOs' demands. This is because of the congestion on transmission line E-D, while C and D are mostly supplied by Solitude with a lower CO_2 rate. Then, the allocated carbon obligation of B is higher than that of C and D, and most of the carbon obligation is assigned to B. For prenucleolus, the excesses in coalitions {B} and {C} are so large that other coalitions' excesses such as those of coalitions {B,D} and {C,D} cannot be reflected in prenucleolus. Then, the assigned carbon obligations of B and C are almost the same, while their different locational factors are not taken into account.

Using FCI, locational demand factors are overestimated while the DSO interactions are not considered. Accordingly, an excessive level of carbon obligation is assigned to D because it is near Brighton with the highest CO_2 rate; the assigned carbon obligation of C depends highly on the status of Solitude. The proportional allocation does not consider the electricity network topology, which results in the same assigned carbon obligation of B and C, though their network locations are different. In addition, the sum of carbon obligations when the three DSOs are supplied separately is smaller than the total carbon obligation of the grand coalition {B,C,D}. Thus, the subadditivity of the game is not satisfied.

Comparatively, only the Shapley value solution evaluates the impacts of DSO locations and interactions appropriately. The difference between the assigned carbon obligations of B and C reflects their congested network locations. The allocated carbon obligation of D is smaller than the FCI allocation and larger than the proportional principle allocation, implying the impacts of DSO locations and interactions are both considered. D with the highest carbon intensity would be more motivated to accommodate the additional renewable generation. Furthermore, a lower D demand will lead to a significant decrease in the Brighton dispatch with the highest CO_2 rate.

C. Case Study 3: IEEE-30-Bus Power System

In this case, a larger power system is tested to show the limitation of the Shapley value-based method for achieving computational feasibility. The IEEE-30-bus power system is depicted in Fig. 3 with the data for generators and DSO demands given in Tables IX and X, respectively [39]. The corresponding generation dispatches are shown in Table XII.

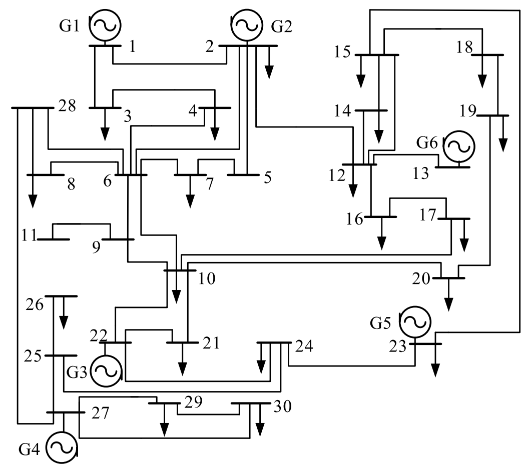

Fig. 3. IEEE-30-bus power system.

TABLE X
GENERATION INFORMATION OF IEEE-30-BUS POWER SYSTEM

Generator	G1	G2	G3	G4	G5	G6
Bus	1	2	22	27	23	13
Energy type	Coal	Coal	Coal	Gas	Gas	Gas
Offer Price ($/MW)	10.00	17.50	20.00	32.50	30.00	30.00
CO_2 rate (t/MW)	0.90	0.80	0.80	0.20	0.30	0.30
Capacity (MW)	80.00	80.00	50.00	55.00	30.00	40.00

TABLE XI
DSO DEMAND INFORMATION OF IEEE-30-BUS POWER SYSTEM (MW)

DSO	DSO1	DSO2	DSO3	DSO4	DSO5	DSO6	DSO7
Bus	2	3	4	7	8	10	12
Demand	21.70	2.40	7.60	22.80	30.00	5.80	11.20
DSO	DSO8	DSO9	DSO10	DSO11	DSO12	DSO13	DSO14
Bus	14	15	16	17	18	19	20
Demand	6.20	8.20	3.50	9.00	3.20	9.50	2.20
DSO	DSO15	DSO16	DSO17	DSO18	DSO19	DSO20	
Bus	21	23	24	26	29	30	
Demand	17.50	3.20	8.70	3.50	2.40	10.60	

TABLE XII
DISPATCH RESULTS OF GENERATORS (MW)

Generator	G1	G2	G3	G4	G5	G6
Power output	80.00	80.00	22.73	0.00	1.51	0.00

Here, the total number of DSO permutations is 20! and the number of non-empty coalitions is $2^{20} - 1$. Since the calculation of the Shapley value-based solution requires $2^{20}-1$ SCED solutions in (1), the Shapley value-based solution becomes computationally infeasible. Here, the Aumann-Shapley rule-based method offers an alternative with the corresponding carbon obligation allocations shown in Table XIII. Accordingly, DSOs with larger demands (e.g., DSO1, DSO4 and DSO5) are assigned a higher carbon obligation, indicating that the DSO's

TABLE XIII
CARBON OBLIGATION ALLOCATION USING AUMANN-SHAPLEY RULE (T CO_2)

DSO	DSO1	DSO2	DSO3	DSO4	DSO5	DSO6	DSO7
Obligation	18.22	2.02	6.38	19.15	20.99	4.88	9.40
DSO	DSO8	DSO9	DSO10	DSO11	DSO12	DSO13	DSO14
Obligation	5.20	6.87	2.94	7.57	2.69	7.98	1.85
DSO	DSO15	DSO16	DSO17	DSO18	DSO19	DSO20	
Obligation	14.75	2.67	7.23	2.92	2.01	8.86	

demand is taken into account. In addition, the assigned carbon obligation of DSO2 is larger than that of DSO16 though they have the same demand, implying the impact of locational factors.

VI. CONCLUSION

Electricity consumption can play a major role in carbon abatement policies and further exploit the carbon reduction potentials in power system operations. In this paper, the system carbon obligation is considered to be allocated among DSOs for motivating DSOs to accommodate more local renewable generations and achieving low-carbon power system operations. The allocation problem is formulated as a cooperative game by considering all DSOs as market players. Several allocation problem constraints including, efficiency, reasonability, symmetry, BC, HMRGC are formulated and analyzed. Several allocation methods are applied in this paper and compared to find if the listed constraints are satisfied in the proposed allocation problem. The results point out that only the Shapley value-based solution can satisfy all the listed constraints. As demonstrated in our case studies, the Shapley value-based solution reflects locational factors and interactions of DSOs properly so that the contribution of each DSO to the total system carbon obligation can be measured. Such allocations can motivate the DSO to accommodate more local renewable generation in the vicinity of thermal units with higher CO_2 rates. The power dispatch of such thermal units will be reduced more significantly as illustrated in case studies. The computational complexities of Shapley value and prenucleolus grow rapidly when the system scales up, since the number of nonempty coalitions grows exponentially with the number of DSO participants. Thus, the Aumann-Shapley rule-based method which is the continuous generalization of the Shapley value satisfies the proposed efficiency, reasonability and symmetry constraints, and provides an alternative solution when involving a large number of DSOs in electricity markets.

REFERENCES

[1] M. Shahidehpour, Z. Li, S. Bahramirad, Z. Li, and W. Tian, "Networked microgrids: Exploring the possibilities of the IIT-Bronzeville grid," *IEEE Power Energy Mag.*, vol. 15, no. 4, pp. 63–71, Jul./Aug. 2017.

[2] H. Lotfi and A. Khodaei, "AC versus DC microgrid planning," *IEEE Trans. Smart Grid*, vol. 8, no. 1, pp. 296–304, Jan. 2017.

[3] Q. Zhou, M. Shahidehpour, Z. Li, and X. Xu, "Two-layer control scheme for maintaining the frequency and the optimal economic operation of hybrid AC/DC microgrids," *IEEE Trans. Power Syst.*, vol. 34, no. 1, pp. 64–75, Jan. 2019.

[4] M. Yan *et al.*, "Coordinated regional-district operation of integrated energy systems for resilience enhancement in natural disasters," *IEEE Trans. Smart Grid*, to be published. doi: 10.1109/TSG.2018.2870358.

[5] Q. Zhou, Z. Li, Q. Wu, and M. Shahidehpour, "Two-stage load shedding for secondary control in hierarchical operation of islanded microgrids," *IEEE Trans. Smart Grid*, to be published. doi: 10.1109/TSG.2018.2817738.

[6] A. Khodaei, "Provisional microgrids," *IEEE Trans. Smart Grid*, vol. 6, no. 3, pp. 1107–1115, May 2015.

[7] J. Munksgaard and K. A. Pedersen, "CO_2 accounts for open economies: Producer or consumer responsibility?" *Energy Policy*, vol. 29, no. 4, pp. 327–334, 2001.

[8] P. Bertoldi, N. Labanca, S. Rezessy, S. Steuwer, and V. Oikonomou, "Where to place the saving obligation: Energy end-users or suppliers?" *Energy Policy*, vol. 63, pp. 328–337, Dec. 2013.

[9] C. Kang *et al.*, "Carbon emission flow from generation to demand: A network-based model," *IEEE Trans. Smart Grid*, vol. 6, no. 5, pp. 2386–2394, Sep. 2015.

[10] B. Li, Y. Song, and Z. Hu, "Carbon flow tracing method for assessment of demand side carbon emissions obligation," *IEEE Trans. Sustain. Energy*, vol. 4, no. 4, pp. 1100–1107, Oct. 2013.

[11] Y. Cheng *et al.*, "Modeling carbon emission flow in multiple energy systems," *IEEE Trans. Smart Grid*, to be published. doi: 10.1109/TSG.2018.2830775.

[12] Q. Zhou, T. Sun, T. Ding, and D. Feng, "Application of carbon intensity in generation expansion planning: A comparative study," in *Proc. Power Energy Soc. Gen. Meeting*, Denver, CO, USA, Jul. 2015, pp. 1–5.

[13] H. Farzin, R. Ghorani, M. Fotuhi-Firuzabad, and M. Moeini-Aghtaie, "A market mechanism to quantify emergency energy transactions value in a multi-microgrid system," *IEEE Trans. Sustain. Energy*, vol. 10, no. 1, pp. 426–437, Jan. 2019.

[14] M. Shahidehpour *et al.*, "Robust two-stage regional-district scheduling of multi-carrier energy systems with a large penetration of wind power," *IEEE Trans. Sustain. Energy*, to be published. doi: 10.1109/TSTE.2018.2864296.

[15] Y. Wang, J. Cheng, N. Zhang, and C. Kang, "Automatic and linearized modeling of energy hub and its flexibility analysis," *Appl. Energy*, vol. 211, pp. 705–714, Feb. 2018.

[16] M. Junqueira *et al.*, "An Aumann–Shapley approach to allocate transmission service cost among network users in electricity markets," *IEEE Trans. Power Syst.*, vol. 22, no. 4, pp. 1532–1546, Nov. 2007.

[17] R. Bhakar, V. S. Sriram, N. P. Padhy, and H. O. Gupta, "Probabilistic game approaches for network cost allocation," *IEEE Trans. Power Syst.*, vol. 25, no. 1, pp. 51–58, Feb. 2010.

[18] M. S. S. Rao *et al.*, "Min-max fair power flow tracing for transmission system usage cost allocation: A large system perspective," *IEEE Trans. Power Syst.*, vol. 25, no. 3, pp. 1457–1468, Aug. 2010.

[19] J. Contreras and F. F. Wu, "A kernel-oriented algorithm for transmission expansion planning," *IEEE Trans. Power Syst.*, vol. 15, no. 4, pp. 1434–1440, Nov. 2000.

[20] N. I. Voropai and E. Y. Ivanova, "Shapley game for expansion planning of generating companies at many non-coincident criteria," *IEEE Trans. Power Syst.*, vol. 21, no. 4, pp. 1630–1637, Nov. 2006.

[21] S. Sharma and A. R. Abhyankar, "Loss allocation for weakly meshed distribution system using analytical formulation of Shapley value," *IEEE Trans. Power Syst.*, vol. 32, no. 2, pp. 1369–1377, Mar. 2017.

[22] D. Songhuai, Z. Xinghua, M. Lu, and X. Hui, "A novel nucleolus-based loss allocation method in bilateral electricity markets," *IEEE Trans. Power Syst.*, vol. 21, no. 1, pp. 28–33, Feb. 2006.

[23] L. Han, T. Morstyn, and M. McCulloch, "Incentivizing prosumer coalitions with energy management using cooperative game theory," *IEEE Trans. Power Syst.*, vol. 34, no. 1, pp. 303–313, Jan. 2019.

[24] P. Chakraborty, E. Baeyens, K. Poolla, P. P. Khargonekar, and P. Varaiya, "Sharing storage in a smart grid: A coalitional game approach," *IEEE Trans. Smart Grid*, to be published. doi: 10.1109/TSG.2018.2858206.

[25] H. T. Nguyen and L. B. Le, "Bi-objective-based cost allocation for cooperative demand-side resource aggregators," *IEEE Trans. Smart Grid*, vol. 9, no. 5, pp. 4220–4235, Sep. 2018.

[26] L. Ma, N. Liu, J. Zhang, and L. Wang, "Real-time rolling horizon energy management for the energy-hub-coordinated prosumer community from a cooperative perspective," *IEEE Trans. Power Syst.*, vol. 34, no. 2, pp. 1227–1242, Mar. 2019.

[27] Z. Hu, L. Chen, D. Gan, and D. Chattopadhyay, "Allocation of unit start-up costs using cooperative game theory," *IEEE Trans. Power Syst.*, vol. 21, no. 2, pp. 653–662, May 2006.

[28] E. Faria, L. A. Barroso, R. Kelman, S. Granville, and M. V. Pereira, "Allocation of firm-energy rights among hydro plants: An Aumann–Shapley approach," *IEEE Trans. Power Syst.*, vol. 24, no. 2, pp. 541–551, May 2009.

[29] L. Freire, A. Street, D. A. Lima, and L. A. Barroso, "A hybrid MILP and benders decomposition approach to find the nucleolus quota allocation for a renewable energy portfolio," *IEEE Trans. Power Syst.*, vol. 30, no. 6, pp. 3265–3275, Nov. 2015.

[30] M. Maschler, "The bargaining set, kernel, and nucleolus," in *Handbook of Game Theory With Economic Applications*. Amsterdam, The Netherlands: North-Holland, 1992, pp. 591–667.

[31] S. Hart and A. Mas-Colell, "Potential, value, and consistency," *Econometrica J. Econometric Soc.*, vol. 57, no. 3, pp. 589–614, 1989.

[32] J. L. Hougaard, *An Introduction to Allocation Rules*. Berlin, Germany: Springer, 2009, pp. 87–91.

[33] D. Schmeidler, "The nucleolus of a characteristic function game," *SIAM J. Appl. Math.*, vol. 17, no. 6, pp. 1163–1170, 1969.

[34] L. S. Shapley, "A value for n-person games," in *Contributions to Game Theory*, H. W. Kuhn and A. W. Tucker, Eds. Princeton, NJ, USA: Princeton Univ. Press, 1953, pp. 307–317.

[35] H. P. Young, "Cost allocation," in *Handbook of Game Theory*, vol. 2. Amsterdam, The Netherlands: North-Holland, 1994, pp. 1193–1235.

[36] M. Ban, M. Shahidehpour, J. Yu, and Z. Li, "A cyber-physical energy management system for optimal sizing and operation of networked nanogrids with battery swapping stations," *IEEE Trans. Sustain. Energy*, vol. 10, no. 1, pp. 491–502, Jan. 2019.

[37] M. Ban, J. Yu, M. Shahidehpour, Y. Yao, and D. Guo, "Considering the differentiating health impacts of fuel emission in optimal generation scheduling," *IEEE Trans. Sustain. Energy*, to be published. doi: 10.1109/TSTE.2018.2879566.

[38] F. Li and R. Bo, "Small test systems for power system economic studies," in *Proc. Power Energy Soc. Gen. Meeting*, Providence, RI, USA, Jul. 2010, pp. 1–4.

[39] R. W. Ferrero, S. M. Shahidehpour, and V. C. Ramesh, "Transaction analysis in deregulated power systems using game theory," *IEEE Trans. Power Syst.*, vol. 12, no. 3, pp. 1340–1347, Aug. 1997.

Quan Zhou (S'16) received the B.S. and M.S. degrees from the Department of Electrical Engineering, Shanghai Jiao Tong University, Shanghai, China, in 2011 and 2016, respectively. He is currently pursuing the Ph.D. degree with the Robert W. Galvin Center for Electricity Innovation, Illinois Institute of Technology. His research interests include microgrid control and optimization, and integration of renewable energy.

Mohammad Shahidehpour (F'01) received the Honorary Doctorate degree from the Polytechnic University of Bucharest, Bucharest, Romania, in 2009. He is a University Distinguished Professor, a Bodine Chair Professor, and the Director of the Robert W. Galvin Center for Electricity Innovation, Illinois Institute of Technology. He is a member of the U.S. National Academy of Engineering, and a fellow of the American Association for the Advancement of Science and the National Academy of Inventors.

Tao Sun (S'11) received the B.S. and M.S. degrees in electrical engineering from Shanghai Jiao Tong University, Shanghai, China, in 2015 and 2018, respectively. He is currently pursuing the Ph.D. degree with the Department of Civil and Environmental Engineering, Stanford University, Stanford, CA, USA. His research interests include market issues and optimization of power systems, with a focus on the integration of distributed energy resources.

Donghan Feng (SM'15) received the B.Sc. and Ph.D. degrees from the Department of Electrical Engineering, Zhejiang University, Hangzhou, China, in 2003 and 2008, respectively. He has been with the faculty of Shanghai Jiao Tong University, Shanghai, China, since 2008, where he is currently a Full Professor, and also serves as the Deputy Director of the State Energy Smart Grid Research and Development Center. His research interests include operation and trading strategies in smart energy networks.

Mingyu Yan (S'18) received the B.S. degree from the College of Electrical Engineering, Zhejiang University, Hangzhou, China, in 2016 and the master's degree from the Huazhong University of Science and Technology, Wuhan, China, in 2018. He is currently pursuing the Ph.D. degree with the Robert W. Galvin Center for Electricity Innovation, Illinois Institute of Technology. His research interests include power system operation and planning, multiple energy system integration, and power system reliability.